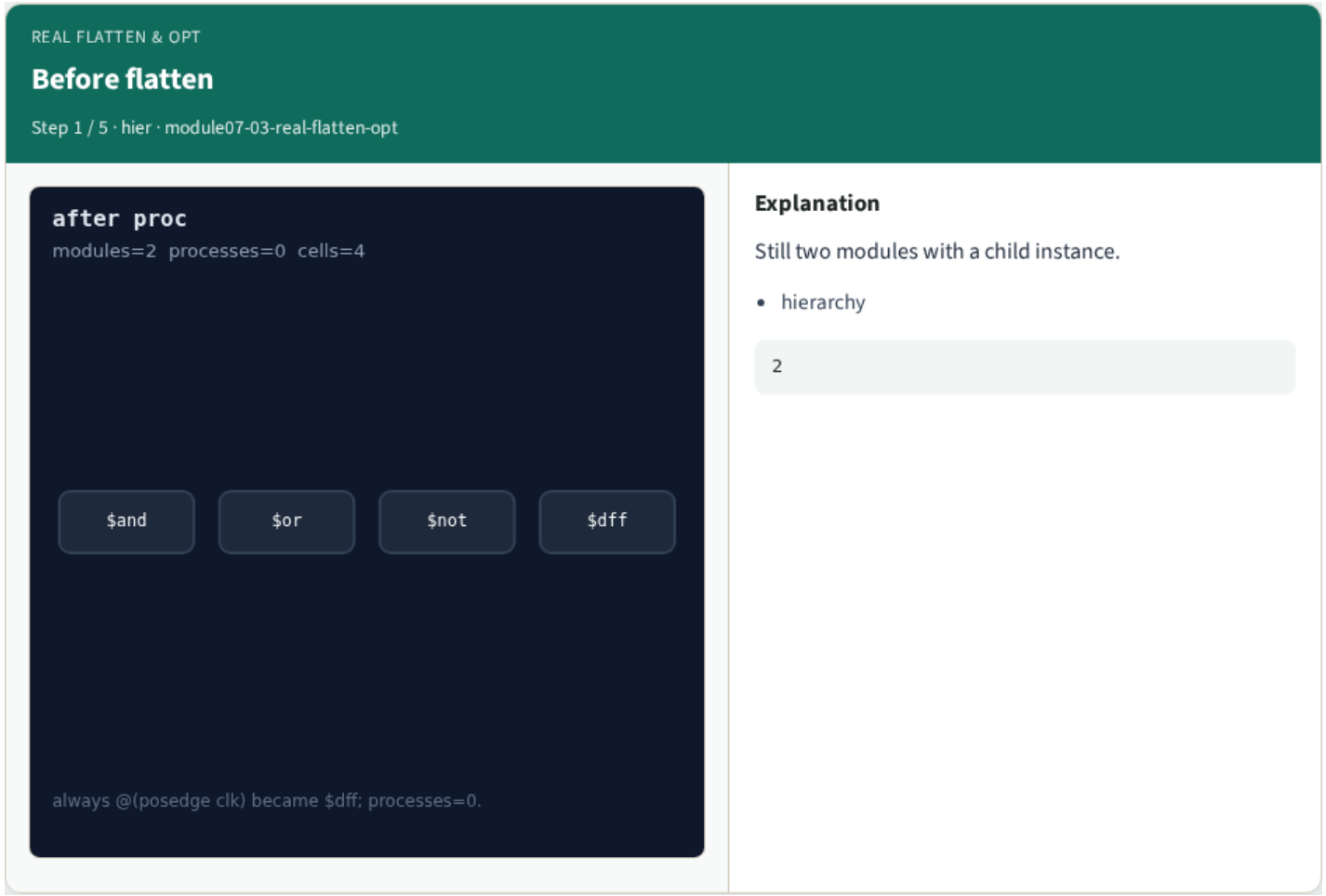
Why real

Flatten inlines child into top; opt merges identical cells and cleans dead logic

The idea

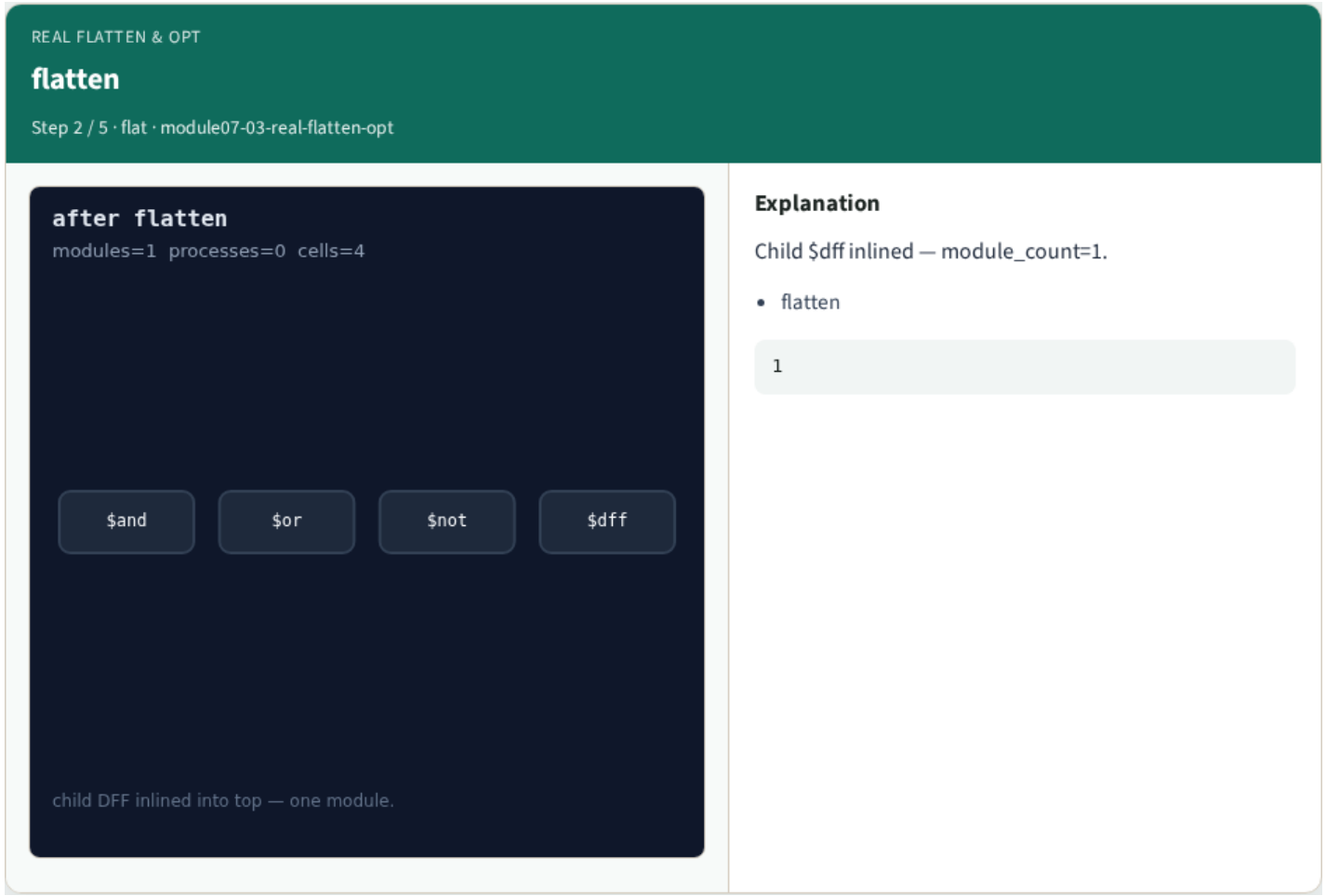
- GOLDEN: flatten modules=1 cells=4; opt cells=4 and_count=1

Before flatten



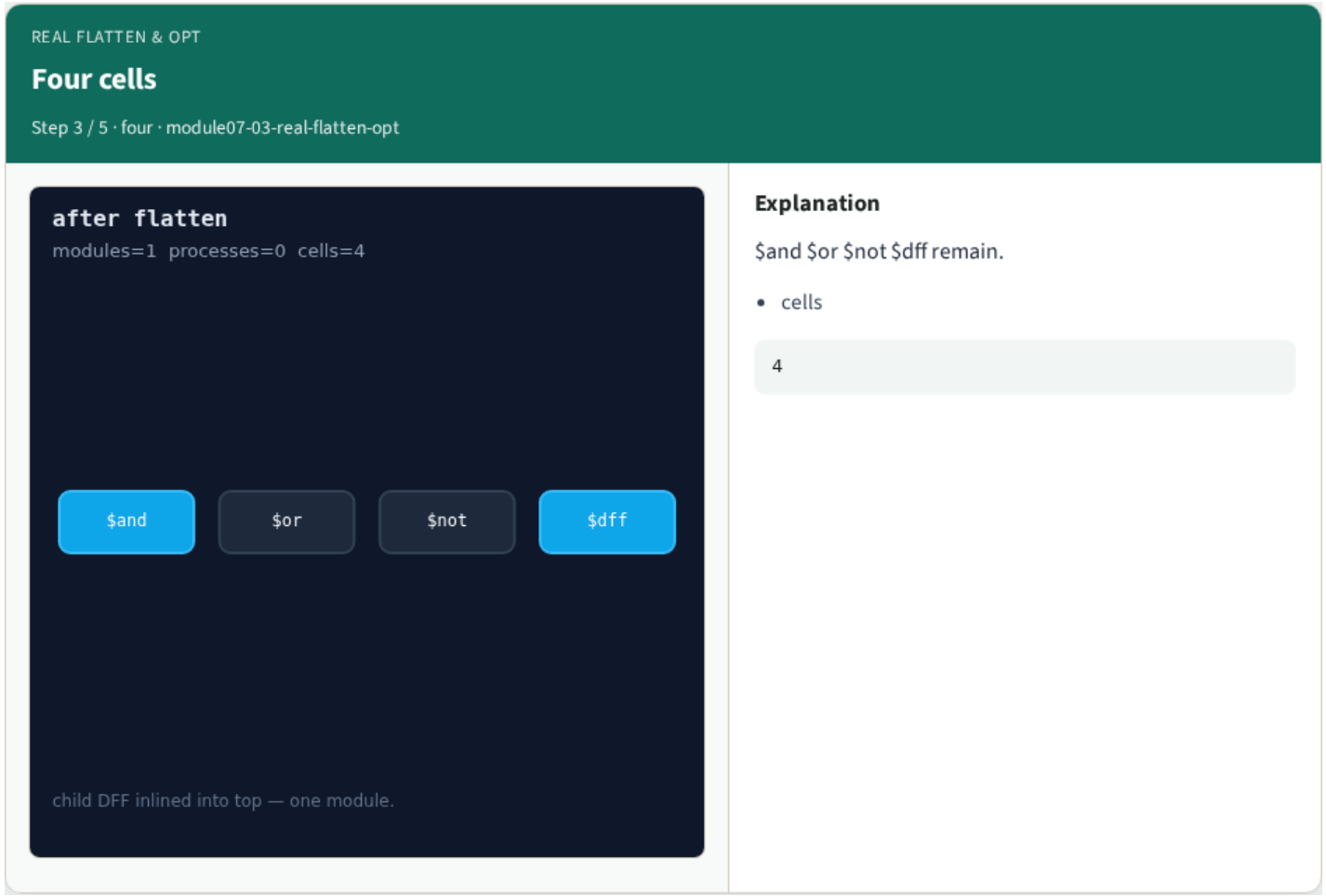
Before flatten

flatten



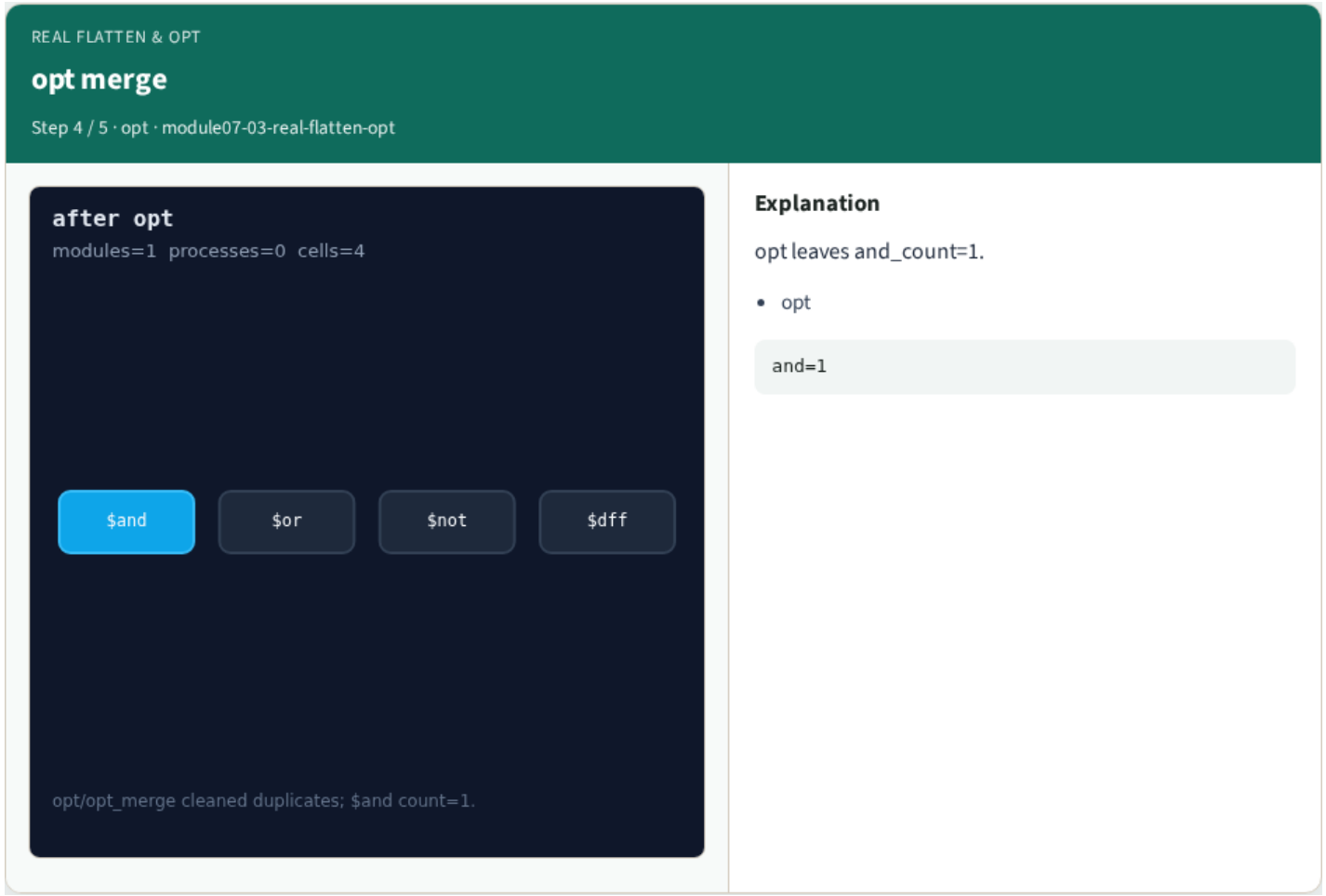
flatten

Four cells



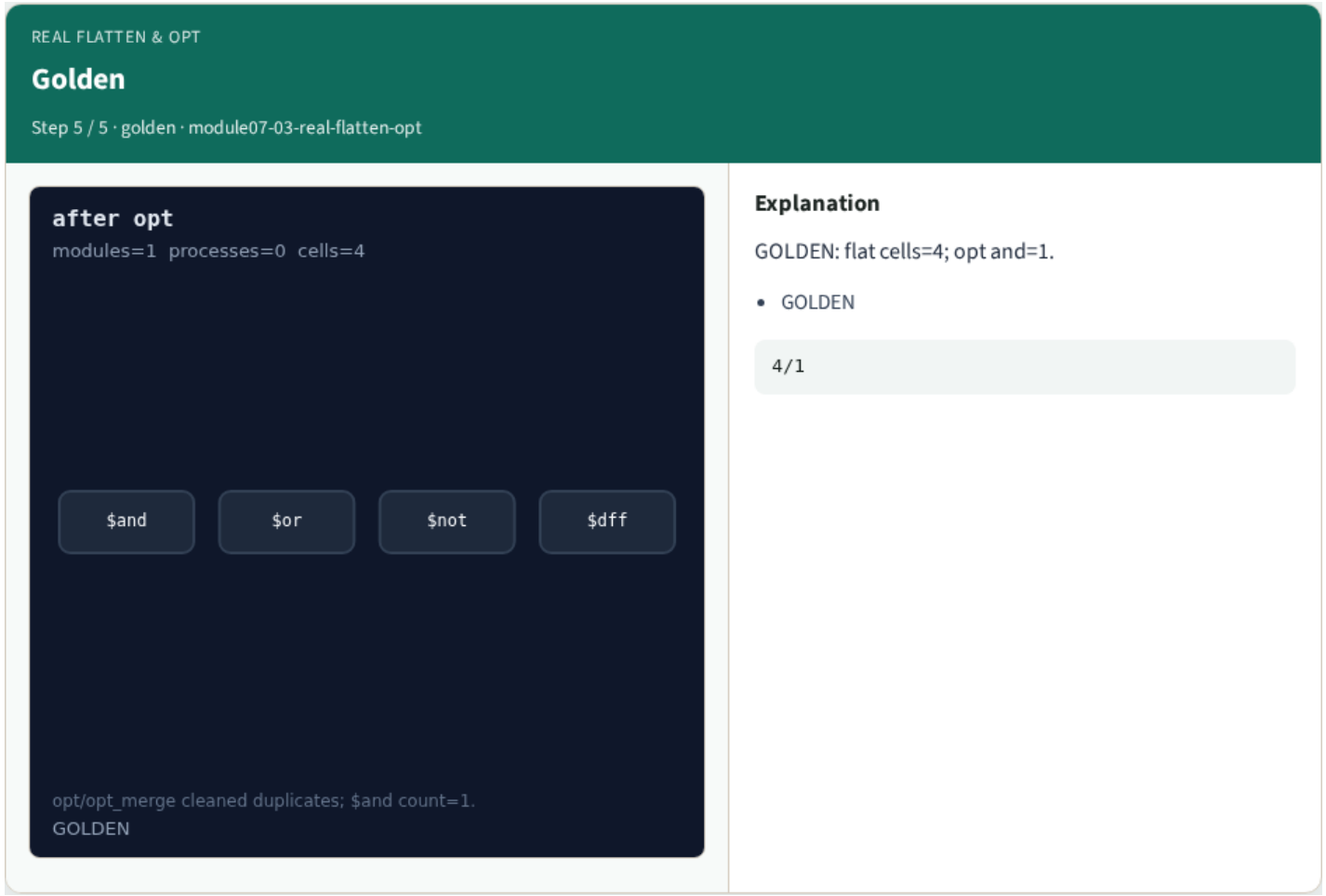
Four cells

opt merge



opt merge

Golden



Golden

Browser lab track

- Open **real-flatten-opt**
- Open **real-flatten-opt**
- Show flatten then opt stages

Implement track

- Call pass_real_flatten_opt(); assert ok

Pitfalls

- Expecting module_count=2 after flatten
- Counting instance cells after flatten

Try these

- From the common folder

Try these — try these

```
cd courses/learn_yosys/common
python3 real_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
flatten then opt
assert modules=1 cells=4 and=1
```

Next

- Complete the quiz for this part
- Next: aigmap + synth